

ECON 600 — Class Notes

What is a Set

Definition (Set). A *set* is a well-defined collection of objects (called *elements*). Sets are written using curly brackets, for example $A = \{a, b, c\}$.

Definition (Membership). If x is an element of A , we write $x \in A$; otherwise $x \notin A$.

Example. Standard number sets

- **Integers:** $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$.
- **Rationals:** $\mathbb{Q} = \{x : x = \frac{g}{r}, g, r \in \mathbb{Z}, r \neq 0\}$.
- **Reals:** $\mathbb{R}$ (not defined here; we use its usual properties).
- **Positive reals:** $\mathbb{R}_{>0} = \{x \in \mathbb{R} : x > 0\}$ (also $\mathbb{R}_{\geq 0} = \{x \in \mathbb{R} : x \geq 0\}$).

Intervals in $\mathbb{R}$

Definition (Intervals). Given $a < b$ in $\mathbb{R}$:

$$(a, b) = \{x \in \mathbb{R} : a < x < b\}, \quad [a, b] = \{x \in \mathbb{R} : a \leq x \leq b\}.$$

Elements and the Empty Set

Example. If $A = \{a, b, c\}$, then a is an element of A , that is $a \in A$.

Definition (Empty set). The *empty set* is the set with no elements. It is denoted $\emptyset$.

Fundamental Property: Order and Repetition Do Not Matter

Remark. A set is determined only by which elements it contains; order and repetition do not change the set. For example

$$\{a, b, c\} = \{a, c, b\} = \{a, a, b, c\}.$$

Equality of Sets

Definition (Equality). We say $X = Y$ if and only if, for every x , $x \in X$ if and only if $x \in Y$.

Proposition (Extensionality). If $X \subseteq Y$ and $Y \subseteq X$, then $X = Y$.

Subsets

Definition (Subset). X is a *subset* of Y (denoted $X \subseteq Y$) if, for every $x \in X$, we have $x \in Y$.

Example (Quick examples).

$$\{a, b\} \subseteq \{a, b, c\}, \quad \mathbb{Z} \subseteq \mathbb{R}, \quad \mathbb{Q} \subseteq \mathbb{R}, \quad \mathbb{R}_{>0} \subseteq \mathbb{R}.$$

Definition (Proper subset). We say X is a *proper subset* of Y if $X \subseteq Y$ but $X \neq Y$. Notation: $X \subsetneq Y$.

Remark. For every set X , we have $\emptyset \subseteq X$ and $X \subseteq X$.

Power Set

Definition (Power set). The *power set* of X is the set of *all* subsets of X . It is denoted 2^X .

Example. If $A = \{a, b, c\}$, then

$$2^A = \{\emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, A\}.$$

Remark. If A is finite with $|A| = n$, then $|2^A| = 2^n$ (useful for counting subsets).

Summary of Notation

- $x \in X$ / $x \notin X$: x belongs / does not belong to X .
- $X \subseteq Y$ / $X \subsetneq Y$: X is a subset (not necessarily proper) / a *proper* subset of Y .
- $\emptyset$: empty set. 2^X : power set of X .
- Intervals: (a, b) open, $[a, b]$ closed.

Useful Mini-Proofs

- 1) $\emptyset \subseteq X$ for all X . If there were an $x \in \emptyset$ with $x \notin X$, we would have an element of the empty set, which is impossible; thus “for all $x \in \emptyset$, $x \in X$ ” is vacuously true.
 - 2) *Criterion for equality.* If $X \subseteq Y$ and $Y \subseteq X$, then by the definition of equality of sets it follows that $X = Y$.
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Set Operations

Union. Given X and Y ,

$$X \cup Y = \{x : x \in X \text{ or } x \in Y\}.$$

For an indexed family $(X_i)_{i \in I}$,

$$\bigcup_{i \in I} X_i = \{x : \exists i \in I \text{ such that } x \in X_i\}.$$

Intersection. Given X and Y ,

$$X \cap Y = \{x : x \in X \text{ and } x \in Y\}.$$

For a family $(X_i)_{i \in I}$,

$$\bigcap_{i \in I} X_i = \{x : \forall i \in I, x \in X_i\}.$$

Difference and complement. For Y and X , the *difference* is

$$Y \setminus X = \{x : x \in Y \text{ and } x \notin X\}.$$

If we fix a *universe* U with $E \subseteq U$, we define the *complement* of E (in U) as $E^c = U \setminus E$.

Disjoint sets. We say that X and Y are *disjoint* if $X \cap Y = \emptyset$. For a family $(X_i)_{i \in I}$:

- **Pairwise disjoint:** $\forall i \neq j, X_i \cap X_j = \emptyset$.
- **Empty total intersection:** $\bigcap_{i \in I} X_i = \emptyset$ (this condition does *not* necessarily imply pairwise disjointness).

Example (Difference between “pairwise” and “total intersection”). Let $X_1 = \{0, 1, 2\}$, $X_2 = \{1, 3\}$ and $X_3 = \{3, 0\}$. Then

$$X_1 \cap X_2 = \{1\} \neq \emptyset, \quad X_1 \cap X_3 = \{0\} \neq \emptyset, \quad X_2 \cap X_3 = \{3\} \neq \emptyset,$$

but

$$\bigcap_{i=1}^3 X_i = \emptyset.$$

The family is *not* pairwise disjoint, even though the total intersection is empty.

Partitions

Definition (Partition). Let X be a set. A collection $\mathcal{P} \subseteq 2^X$ is a *partition* of X if:

- (i) $E \neq \emptyset$ for every $E \in \mathcal{P}$ (**nonempty blocks**);
- (ii) If $E, F \in \mathcal{P}$ and $E \cap F \neq \emptyset$, then $E = F$ (**pairwise disjoint**);
- (iii) $\bigcup_{E \in \mathcal{P}} E = X$ (**covering**).

The elements of $\mathcal{P}$ are called *classes* or *blocks*.

Remark. A partition decomposes X into disjoint pieces that cover it entirely. Each $x \in X$ belongs to a unique block.

Cartesian Product

Definition (Cartesian product of two sets). For X and Y ,

$$X \times Y = \{(x, y) : x \in X, y \in Y\}.$$

Order matters: in general $(x, y) \neq (y, x)$.

Definition (Cartesian product of a family). Given a family $(X_i)_{i \in I}$,

$$\prod_{i \in I} X_i = \{(x_i)_{i \in I} : \forall i \in I, x_i \in X_i\}.$$

Example. Let $C = \{1, 2, \dots, c\}$ and let $\mathbb{R}_{>0}$ be the positive reals. Then

$$C \times \mathbb{R}_{>0} = \{(c, w) : c \in C, w \in \mathbb{R}_{>0}\}.$$

Example. If $X_1 = \{1, 2\}$ and $X_2 = \{\alpha, \beta\}$, then

$$X_1 \times X_2 = \{(1, \alpha), (1, \beta), (2, \alpha), (2, \beta)\}.$$

Definition (n -fold product). For a set X and $n \in \mathbb{N}$,

$$X^n = \underbrace{X \times X \times \cdots \times X}_{n \text{ factors}}.$$

Examples: $\mathbb{R}^2, \mathbb{R}^n$.

Relations

Definition (Binary relation). Given sets X and Y , a *relation* from X to Y is a subset $R \subseteq X \times Y$. We write $x R y$ to indicate $(x, y) \in R$. Formally:

$$R = \{(x, y) \in X \times Y : \varphi(x, y)\}$$

where $\varphi(x, y)$ is a property/predicate that decides when (x, y) is “related.”

Definition (Domain, image, and inverse relation). For $R \subseteq X \times Y$,

$$\text{Dom}(R) = \{x \in X : \exists y \in Y \text{ with } (x, y) \in R\}, \quad \text{Im}(R) = \{y \in Y : \exists x \in X \text{ with } (x, y) \in R\},$$

and the *inverse relation* is $R^{-1} = \{(y, x) \in Y \times X : (x, y) \in R\}$.

Remark. When $X = Y$, we speak of a relation *on* X ; in that case, properties such as *reflexivity*, *symmetry*, *antisymmetry*, and *transitivity* are often of interest (to be introduced in the next class).

Minimal set algebra (useful). For $A, B, C \subseteq U$:

$$(A \cup B)^c = A^c \cap B^c, \quad (A \cap B)^c = A^c \cup B^c, \quad A \setminus B = A \cap B^c.$$

“if and only if” statements are proved by double inclusion.

Properties of Relations in $X \times X$

Definition (Relation). A relation R between X and Y is a subset $R \subseteq X \times Y$. We write $x R y$ for $(x, y) \in R$. In what follows, $R \subseteq X \times X$.¹

Definition. Basic properties

- **Reflexive:** $\forall x \in X, x R x$.
- **Irreflexive:** $\forall x \in X, \neg(x R x)$.
- **Symmetric:** $x R y \Rightarrow y R x$.
- **Antisymmetric:** $(x R y \ \& \ y R x) \Rightarrow x = y$.
- **Asymmetric:** $x R y \Rightarrow \neg(y R x)$ (implies irreflexive and antisymmetric).
- **Transitive:** $(x R y \ \& \ y R z) \Rightarrow x R z$.
- **Complete (or connected):** $\forall x, y \in X, x R y \text{ or } y R x$.

Example A: $\geq$ on $\mathbb{R}$. Define $x \geq y \iff x$ is greater than or equal to y .

- Reflexive: $x \geq x$.
- Transitive: $x \geq y$ and $y \geq z \Rightarrow x \geq z$.
- Antisymmetric: $x \geq y$ and $y \geq x \Rightarrow x = y$.
- Not symmetric: $3 \geq 2$ but $2 \not\geq 3$.
- Complete: for all x, y , either $x \geq y$ or $y \geq x$.

¹It is useful to interpret a relation as a formal device for comparing two objects. If $R \subseteq X \times X$, then an element of R is an ordered pair (x, y) , where $x, y \in X$, and $x R y$ means that x stands in the relationship R to y . This distinction is particularly useful in consumer theory. There, one typically has $X \subseteq \mathbb{R}^n$, so that a single element $x = (x_1, \dots, x_n) \in X$ is an entire consumption bundle. The relation then compares two such bundles: $(x, y) \in X \times X$. For instance, if $R = \succeq$ is a weak preference relation, $x \succeq y$ means that bundle x is weakly preferred to bundle y . More generally, this is the basic language used to represent preferences over alternatives, including in social choice theory.

Example B: $>$ on $\mathbb{R}$. Define $x > y \iff x$ is strictly greater than y .

- Not reflexive (indeed *irreflexive*): $x \not> x$.
- Transitive: if $x > y$ and $y > z$, then $x > z$.
- **Asymmetric** $\Rightarrow$ **antisymmetric** vacuously: if $x > y$, then *never* $y > x$ (no pairs go both directions).
- Not complete: when $x = y$, neither $x > y$ nor $y > x$.

Margin marks:

- “think a bit more”: the statement “it is antisymmetric because there are no x, y with $x > y$ and $y > x$ ” is correct: the implication of antisymmetry holds vacuously (no counterexamples).
- “not complete because it’s not reflexive”: this is a good intuition and indeed a **theorem**:

Proposition (Completeness implies reflexivity). *If R is complete, then it is reflexive.*

Proof. Fix $x \in X$. By completeness applied to the pair (x, x) we must have xRx . □

Remark. The *converse* is false: a relation may be reflexive and yet *not* complete (e.g. equality = on X).

Example (product order on $\mathbb{R}^2$). Define for $x = (x_1, x_2)$, $y = (y_1, y_2)$:

$$x \succeq y \iff (x_1 \geq y_1 \text{ and } x_2 \geq y_2).$$

Then $\succeq$ is **reflexive**, **transitive**, and **antisymmetric** (a *partial order*), but *not* complete.

Example. $(2, 1)$ and $(1, 10)$ are **incomparable**: neither $(2, 1) \succeq (1, 10)$ (since $1 \not\geq 10$) nor $(1, 10) \succeq (2, 1)$ (since $1 \not\geq 2$).

Equivalence and Indifference Relations

Definition (Equivalence). A relation E on X is an *equivalence* if it is **reflexive**, **symmetric**, and **transitive**. Its equivalence classes form a partition of X .

Example D (identity). Let $X = \{a, b, c\}$ and $E = \{(a, a), (b, b), (c, c)\}$. Then E is **reflexive**, **symmetric**, and **transitive**. *On your orange note (“I don’t understand transitivity”):* transitivity requires: if xEy and yEz , then xEz . Here the only possible chains are $xEEx$ (with $x = a, b$ or c), so $xEEx$ holds; there are no “mixed” chains, hence the condition is true. Moreover, E is *not* complete (e.g. neither aEb nor bEa).

Indifference induced by a weak preference. Let $\succeq$ be a **reflexive** and **transitive** relation on X (a weak preference). Define

$$x \sim y \iff (x \succeq y \text{ and } y \succeq x).$$

Proposition. *The relation $\sim$ is an **equivalence** (reflexive, symmetric, and transitive).*

Proof. Reflexive: $x \succeq x$ gives $x \sim x$. Symmetric: the definition is commutative. Transitive: if $x \sim y$ and $y \sim z$, then $x \succeq y \succeq z$ and $z \succeq y \succeq x$; by transitivity of $\succeq$, $x \succeq z$ and $z \succeq x$, hence $x \sim z$. □

Guided summary (connection with your notes).

- $>$: irreflexive, asymmetric, transitive, *not* complete. Antisymmetry holds vacuously.
- $\geq$: total order (reflexive, antisymmetric, transitive, complete), but not symmetric.
- Completeness $\Rightarrow$ reflexivity; *not* conversely (e.g. = or product order on $\mathbb{R}^2$).
- Identity and “indifference” are equivalence relations; equivalence does not require completeness.

Functions

Function as a Special Relation

Definition (Function). Let X and Y be sets. A *function* f from X to Y is a relation $G \subseteq X \times Y$ that satisfies:

- (i) **Existence:** for every $x \in X$ there exists $y \in Y$ such that $(x, y) \in G$;
- (ii) **Uniqueness:** if $(x, y) \in G$ and $(x, y') \in G$, then $y = y'$.

In that case we write $f : X \rightarrow Y$ and $f(x) = y$ when $(x, y) \in G$. The set G is called the *graph* of f and is denoted

$$\text{Gr}(f) = \{(x, y) \in X \times Y : y = f(x)\}.$$

Remark (On your orange note: “Is this the same as reflexive?”). No. **Reflexivity** of a relation on X says $(x, x) \in R$ for every $x \in X$, that is, “each x is related to itself.” Here condition (i) is the *existence of some image* y for each x , and it need not be the case that $y = x$.

Domains, Codomains, Image (and the Word “Range”)

Definition. For $f : X \rightarrow Y$:

- X is the **domain**;
- Y is the **codomain**;
- $\text{Im}(f) = f(X) = \{f(x) : x \in X\} \subseteq Y$ is the **image** (the values actually attained).

Remark (Your green note). Some authors use “*range*” to mean the *codomain* and others to mean the *image*. To avoid ambiguity, in this course we adopt: *codomain* = Y , *image* = $f(X)$.

Two Baseline Examples from Class

Example (Finite domain). Let $X = \{-9, 200\}$, $Y = \mathbb{R}$, and $f(-9) = 10$, $f(200) = -1$. Then

$$\text{Gr}(f) = \{(-9, 10), (200, -1)\}, \quad \text{Im}(f) = \{10, -1\}.$$

Example (Increasing parabola on $X = \mathbb{R}_{>0}$). Let $f : \mathbb{R}_{>0} \rightarrow \mathbb{R}$, $f(x) = x^2$. For each x there is a unique y (“*vertical line test*”): it is a function. It is **injective** on $\mathbb{R}_{>0}$ (strictly increasing), but it is *not* **surjective** onto $\mathbb{R}$ (it does not take negative values).

Example (Concave parabola $f(x) = ax - x^2$ on $X = \mathbb{R}_{>0}$). The graph passes the vertical line test (it is a function), but it is typically *not* injective: for certain positive y there are two distinct x with $f(x) = y$ (your orange note “*there might be multiple x ’s with $f(x) = y$* ”).

Direct Image and Inverse Image

Definition (Image of a subset). For $E \subseteq X$, the **image of E** under f is

$$f(E) = \{y \in Y : \exists x \in E \text{ with } y = f(x)\} \subseteq Y.$$

Definition (Inverse image of a subset). For $F \subseteq Y$, the **inverse image** (or *preimage*) is

$$f^{-1}(F) = \{x \in X : f(x) \in F\} \subseteq X.$$

In particular, $f^{-1}(\{y\})$ (the *fiber* over y) is the set of all x that map to y .

Example (Your example 10b). If $f(x) = ax - x^2$ and y^* lies below the maximum of the parabola, then

$$f^{-1}(\{y^*\}) = \{x_1, x_2\} \quad \text{with } x_1 \neq x_2.$$

This shows that the *inverse image* is a *set*; it is **not**, in general, a *functional inverse*.

Injectivity, Surjectivity, Bijectivity

Definition (Types of functions). Let $f : X \rightarrow Y$.

- **Injective** (one-to-one): for all $x_1 \neq x_2$, $f(x_1) \neq f(x_2)$. Equivalent formulation: $f(x_1) = f(x_2) \Rightarrow x_1 = x_2$. Equivalent in terms of fibers: $|f^{-1}(\{y\})| \leq 1$ for every y .
- **Surjective** (onto): $\text{Im}(f) = Y$. Equivalent in terms of fibers: $f^{-1}(\{y\}) \neq \emptyset$ for every $y \in Y$.
- **Bijective**: both injective and surjective.

Inverse Function

Proposition (Characterization of invertibility). A function $f : X \rightarrow Y$ is **invertible** if and only if it is **bijective**. In that case there exists a unique function $g : Y \rightarrow X$ (the inverse of f) such that

$$g(f(x)) = x \text{ for every } x \in X, \quad \text{and} \quad f(g(y)) = y \text{ for every } y \in Y.$$

Proof idea. If f is bijective, then for each $y \in Y$ there is exactly one $x \in X$ with $f(x) = y$; defining $g(y) = x$ satisfies the identities. Conversely, if there exists g with $g \circ f = \text{id}_X$ and $f \circ g = \text{id}_Y$, then each y equals $f(x)$ for a unique x , hence f is bijective. $\square$

Notation and Graphical “Tests”

- $f : X \rightarrow Y$ denotes the function; $f(x)$ denotes its *value* at x .
- $\mathcal{F}(X, Y)$: the set of all functions from X to Y .
- **Vertical line test**: a curve in the plane describes a function $x \mapsto y$ if *every* vertical line intersects the curve in at most one point (your orange note “for each x there is a unique y ”).
- **Horizontal line test**: a function is injective if *every* horizontal line intersects the graph in at most one point (your orange note “there may exist multiple x with the same y ” indicates failure of injectivity).

Types of Functions

Let $f : X \rightarrow Y$.

- **Injective** (one-to-one): if $x_1 \neq x_2$ then $f(x_1) \neq f(x_2)$.
- **Surjective** (onto): for every $y \in Y$ there exists $x \in X$ such that $f(x) = y$ (equivalently: $f(X) = Y$).
- **Bijective**: both injective and surjective (equivalently: invertible).

Basic Rules for Images and Preimages

Proposition (Monotonicity and inclusions). Let $f : X \rightarrow Y$. Then:

- (1) If $E \subseteq E' \subseteq X$, then $f(E) \subseteq f(E')$.
- (2) If $F \subseteq F' \subseteq Y$, then $f^{-1}(F) \subseteq f^{-1}(F')$.
- (3) For every $E \subseteq X$, one has $E \subseteq f^{-1}(f(E))$.
- (4) For every $F \subseteq Y$, one has $f(f^{-1}(F)) \subseteq F$.

Proof. All proofs proceed by *element chasing*.

(1) Let $y \in f(E)$. By definition, there exists $x \in E$ with $f(x) = y$. Since $E \subseteq E'$, then $x \in E'$, hence $y = f(x) \in f(E')$. Therefore $f(E) \subseteq f(E')$.

(2) Let $x \in f^{-1}(F)$. By definition, $f(x) \in F$. Since $F \subseteq F'$, we also have $f(x) \in F'$, so $x \in f^{-1}(F')$. Therefore $f^{-1}(F) \subseteq f^{-1}(F')$.

(3) Let $x \in E$. Then $f(x) \in f(E)$ by definition of image. Hence $x \in f^{-1}(f(E))$. Thus $E \subseteq f^{-1}(f(E))$.

(4) Let $y \in f(f^{-1}(F))$. Then there exists $x \in f^{-1}(F)$ with $f(x) = y$. But $x \in f^{-1}(F)$ means $f(x) \in F$, that is, $y \in F$. Hence $f(f^{-1}(F)) \subseteq F$. $\square$

Important remarks (when equalities may fail).

- **(3) may be strict if f is not injective.** Classic example: $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x^2$. Take $E = \{1\}$. Then $f(E) = \{1\}$ and

$$f^{-1}(f(E)) = f^{-1}(\{1\}) = \{-1, 1\} \supsetneq E.$$

This matches your sketch of the concave parabola: a single y -value may correspond to multiple x 's in the preimage.

- **(4) may be strict if F contains values outside $\text{Im}(f)$.** Again with $f(x) = x^2$, take $F = (-1, 1)$. Then $f^{-1}(F) = (-1, 1)$ and

$$f(f^{-1}(F)) = f((-1, 1)) = [0, 1) \subsetneq (-1, 1) = F.$$

The intuition: f does not “reach” negative values, so it cannot recover all of F .

- **Useful characterizations.**

a) f is **injective** $\iff$ for every $E \subseteq X$ one has $f^{-1}(f(E)) = E$.

b) f is **surjective** $\iff$ for every $F \subseteq Y$ one has $f(f^{-1}(F)) = F$.

Proofs: the forward direction follows from (3) and (4) plus the definition of injectivity/surjectivity; the reverse direction follows by taking $E = \{x\}$ in a) and $F = \{y\}$ in b).

Distribution of preimage and image over unions and intersections. For $F, G \subseteq Y$ and $E, H \subseteq X$:

$$f^{-1}(F \cup G) = f^{-1}(F) \cup f^{-1}(G), \quad f^{-1}(F \cap G) = f^{-1}(F) \cap f^{-1}(G).$$

For images:

$$f(E \cup H) = f(E) \cup f(H), \quad f(E \cap H) \subseteq f(E) \cap f(H) \quad (\text{equality may fail if } f \text{ is not injective}).$$

When (3) and (4) are Equalities

Corollary. Let $f : X \rightarrow Y$.

- a) f is **injective** $\iff$ for every $E \subseteq X$ one has

$$f^{-1}(f(E)) = E.$$

- b) f is **surjective** $\iff$ for every $F \subseteq Y$ one has

$$f(f^{-1}(F)) = F.$$

Counterexamples.

- **(3) may be strict if f is not injective.** With $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x^2$ and $E = \{1\}$:

$$f(E) = \{1\}, \quad f^{-1}(f(E)) = \{-1, 1\} \supsetneq E.$$

- **(4) may be strict if f is not surjective.** With $f : \mathbb{R} \rightarrow \{a, b\}$ given by $f(x) = a$ for all $x \in \mathbb{R}$, and $F = \{a, b\} \subseteq Y$:

$$f^{-1}(F) = \mathbb{R}, \quad f(f^{-1}(F)) = f(\mathbb{R}) = \{a\} \subsetneq \{a, b\} = F.$$

Remark. Here $f^{-1}(F)$ is the *inverse image* (preimage) of a set $F \subseteq Y$; it is **not** the “inverse function.” Only when f is bijective does the inverse function $f^{-1} : Y \rightarrow X$ exist, and in that case $f^{-1}(F)$ coincides with the preimage via that inverse function.

Composition of Functions

Let $f : X \rightarrow Y$ and $g : Y \rightarrow Z$. We define the **composition**

$$g \circ f : X \rightarrow Z, \quad (g \circ f)(x) = g(f(x)).$$

Type check. The composition $g \circ f$ is well defined whenever $f(X) \subseteq \text{Dom}(g)$. In the standard formulation $f : X \rightarrow Y$ and $g : Y \rightarrow Z$, this condition is automatically satisfied.

Example (Order matters). Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$, $f(x_1, x_2) = 3x_1 + 2x_1x_2$, and $g : \mathbb{R} \rightarrow \mathbb{R}$, $g(t) = 3t$. Then $g \circ f : \mathbb{R}^2 \rightarrow \mathbb{R}$ is well defined and

$$(g \circ f)(x_1, x_2) = g(3x_1 + 2x_1x_2) = 3[3x_1 + 2x_1x_2].$$

In contrast, $f \circ g$ does *not* make sense because g returns a real while f expects a pair in $\mathbb{R}^2$.

Useful properties.

- **Associativity:** $h \circ (g \circ f) = (h \circ g) \circ f$ (when compositions are well typed).
- **Identity:** $\text{id}_Y \circ f = f$ and $f \circ \text{id}_X = f$.
- **Monotonicity of properties:**
 - If f and g are *injective*, then $g \circ f$ is injective.
 - If f and g are *surjective*, then $g \circ f$ is surjective.
 - If f and g are *bijective*, then $g \circ f$ is bijective; its inverse is $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$.

Correspondences (Multivalued Maps)

Definition (Correspondence). A *correspondence* (or multivalued map) from X to Y is a rule Γ that assigns to each $x \in X$ a nonempty set $\Gamma(x) \subseteq Y$. We can view it as a function

$$\Gamma : X \longrightarrow 2^Y \setminus \{\emptyset\}, \quad x \mapsto \Gamma(x).$$

Its **graph** is $\text{Gr}(\Gamma) = \{(x, y) \in X \times Y : y \in \Gamma(x)\}$.

Remark. A correspondence *is* a function, but its *codomain* is not Y but the *power set* $2^Y \setminus \{\emptyset\}$. It does not require uniqueness of the value: it allows multiple “outputs” for each x .

Example (Your discrete sets). Let $X = \{a, b\}$ and $Y = \{x, y, z, q, r\}$. Define

$$\Gamma(a) = \{x, y, z\}, \quad \Gamma(b) = \{y, q, r\}.$$

Then $\Gamma : X \rightarrow 2^Y \setminus \{\emptyset\}$ is a correspondence with

$$\text{Gr}(\Gamma) = \{(a, x), (a, y), (a, z), (b, y), (b, q), (b, r)\}.$$

Image and preimage for correspondences. For $E \subseteq X$ and $F \subseteq Y$:

$$\Gamma(E) = \bigcup_{x \in E} \Gamma(x) \subseteq Y, \quad \Gamma^{-1}(F) = \{x \in X : \Gamma(x) \cap F \neq \emptyset\} \subseteq X.$$

(Note: in the preimage we require “*nonempty intersection*” with F .)

Definition (Selections). A *selection* of Γ is a function $s : X \rightarrow Y$ such that $s(x) \in \Gamma(x)$ for all x . (Useful when one wants to “pick” an element from each set of the correspondence.)

Real Numbers and Bounds

Upper and Lower Bounds

Let $X \subseteq \mathbb{R}$ be a nonempty set.

Definition. Lower and upper bounds:

- $u \in \mathbb{R}$ is an **upper bound** of X if $\forall x \in X, x \leq u$.
- $\ell \in \mathbb{R}$ is a **lower bound** of X if $\forall x \in X, \ell \leq x$.

Definition (Bounded). We say that X is **bounded above** if it has some upper bound; **bounded below** if it has some lower bound; and **bounded** if it has both.

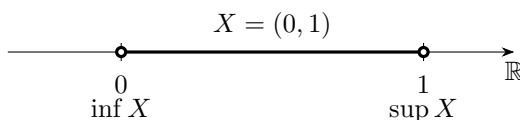
Remark. A set may have neither an upper nor a lower bound (for example, $\mathbb{Z} \subset \mathbb{R}$ has no upper bound).

Supremum and Infimum

Definition (Supremum and infimum). Let $X \subseteq \mathbb{R}$ be nonempty.

- The **supremum** of X , denoted $\sup X$, is the *least* upper bound of X : it is an upper bound and satisfies $\sup X \leq u$ for every upper bound u .
- The **infimum** of X , denoted $\inf X$, is the *greatest* lower bound of X : it is a lower bound and satisfies $\ell \leq \inf X$ for every lower bound ℓ .

Remark. $\sup X$ and $\inf X$ need *not* belong to X . For example, if $X = (0, 1)$, then $\inf X = 0 \notin X$ and $\sup X = 1 \notin X$.



Axiom of Completeness of $\mathbb{R}$

Definition (Axiom of completeness). Every nonempty subset $X \subseteq \mathbb{R}$ that is bounded above has a **supremum** in $\mathbb{R}$; and every nonempty subset that is bounded below has an **infimum** in $\mathbb{R}$.

Remark. This *fails* in $\mathbb{Q}$. For example, $X = \{q \in \mathbb{Q} : q^2 < 2\}$ is nonempty and bounded above, but $\sup X = \sqrt{2} \notin \mathbb{Q}$. This is why we work in $\mathbb{R}$ when using suprema/infima without further hypotheses.

Remark (Convention). If X has no upper bound, we write $\sup X = +\infty$; if it has no lower bound, $\inf X = -\infty$. (These are *conventional* values useful for stating results.)

Basic Properties of the Supremum

Proposition. Let $X \subseteq \mathbb{R}$ be nonempty and bounded above. Then:

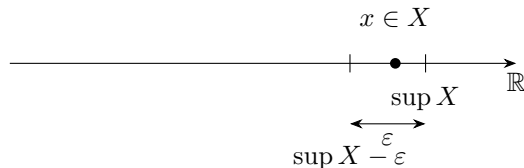
- $\sup X$ is **unique**.
- (Approximation by elements of X) For every $\varepsilon > 0$ there exists $x \in X$ such that

$$\sup X - \varepsilon < x \leq \sup X.$$

Dually, if X is bounded below, then for every $\varepsilon > 0$ there exists $x \in X$ with $\inf X \leq x < \inf X + \varepsilon$.

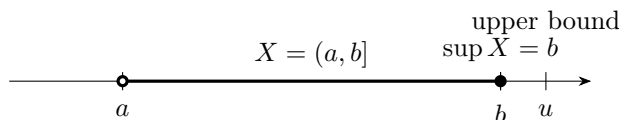
Proof. (i) If u_1 and u_2 are suprema, then $u_1 \leq u_2$ (since u_2 is an upper bound) and $u_2 \leq u_1$ (since u_1 is an upper bound). Hence $u_1 = u_2$.

(ii) Let $\varepsilon > 0$. Suppose there is *no* $x \in X$ with $\sup X - \varepsilon < x$. Equivalently, $\forall x \in X, x \leq \sup X - \varepsilon$. But then $\sup X - \varepsilon$ would be an upper bound of X *smaller* than $\sup X$, contradicting minimality of $\sup X$. Therefore, there exists $x \in X$ with $\sup X - \varepsilon < x$. Since $\sup X$ is an upper bound, we also have $x \leq \sup X$. $\square$

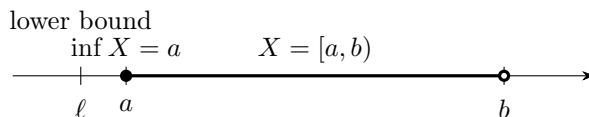


Two Conceptual Diagrams

Upper bounds and supremum.



Lower bounds and infimum (dual).



Alternative Proofs for Prop. (ii)

Recall: if $X \neq \emptyset$ is bounded above, then for every $\varepsilon > 0$ there exists $x \in X$ such that $\sup X - \varepsilon < x \leq \sup X$.

Proof. (a) Proof by contradiction. Suppose there exists $\varepsilon > 0$ such that $\forall x \in X, x \leq \sup X - \varepsilon$. Then $\sup X - \varepsilon$ is an *upper bound* of X and is strictly smaller than $\sup X$, contradicting the minimality of $\sup X$. $\square$

Proof. (b) Proof by contrapositive. Let u be an upper bound of X . If there exists $\varepsilon > 0$ such that $u - \varepsilon \geq x$ for all $x \in X$, then $u - \varepsilon$ would also be an upper bound, hence u could not be the *least* upper bound. By contrapositive, if $u = \sup X$, then for every $\varepsilon > 0$ there must exist $x \in X$ with $x > u - \varepsilon$. $\square$

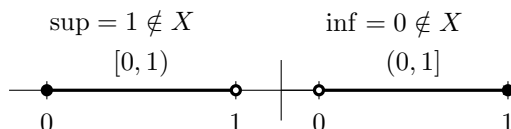
Maximum and Minimum

Definition (Maximum and minimum). Let $X \subseteq \mathbb{R}$ be nonempty.

- $\bar{x}$ is a **maximum** of X if $\bar{x} \in X$ and $x \leq \bar{x}$ for every $x \in X$.
- $\underline{x}$ is a **minimum** of X if $\underline{x} \in X$ and $\underline{x} \leq x$ for every $x \in X$.

Remark. There may be bounds without a maximum/minimum. Examples:

$$X = [0, 1) \quad (\text{no maximum}), \quad X = (0, 1] \quad (\text{no minimum}).$$



Proposition (Maximum/minimum vs. supremum/infimum). Let $X \subseteq \mathbb{R}$ be nonempty.

(i) $\bar{x}$ is a **maximum** of $X \iff \bar{x} = \sup X$ and $\bar{x} \in X$.

(ii) $\underline{x}$ is a **minimum** of $X \iff \underline{x} = \inf X$ and $\underline{x} \in X$.

Proof. (i) If $\bar{x}$ is a maximum, then it is an upper bound and belongs to X ; by minimality of the supremum, $\sup X \leq \bar{x}$, and since $\bar{x} \in X$ is already a bound, then $\bar{x} \leq \sup X$, so $\sup X = \bar{x}$. Conversely, if $\bar{x} = \sup X \in X$, then no $x \in X$ exceeds $\bar{x}$; hence it is a maximum. The case (ii) is dual. $\square$

Corollary. X has a **maximum** $\iff \sup X \in X$ (in that case, it is unique). Similarly, X has a **minimum** $\iff \inf X \in X$.

Metric Spaces

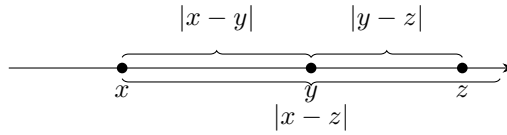
Definition (Metric). Let X be a set. A **metric** on X is a function $d : X \times X \rightarrow \mathbb{R}$ such that, for all $x, y, z \in X$:

- (a) $d(x, y) \geq 0$ (nonnegativity);
- (b) $d(x, y) = 0 \iff x = y$ (identity of indiscernibles);
- (c) $d(x, y) = d(y, x)$ (symmetry);
- (d) $d(x, z) \leq d(x, y) + d(y, z)$ (triangle inequality).

Definition (Metric space). A **metric space** is a pair (X, d) with $X \neq \emptyset$ and d a metric on X .

Example 1: usual metric on $\mathbb{R}$. For $X \subseteq \mathbb{R}$, $d_u(x, y) = |x - y|$. Properties (a)–(c) follow from properties of the absolute value. For (d):

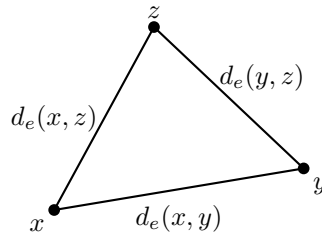
$$|x - z| = |(x - y) + (y - z)| \leq |x - y| + |y - z|.$$



Example 2: Euclidean metric on $\mathbb{R}^n$. For $X \subseteq \mathbb{R}^n$, $x = (x_1, \dots, x_n)$, $y = (y_1, \dots, y_n)$,

$$d_e(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}.$$

When $n = 1$, $d_e = d_u$.



Example 3: Discrete Metric

Let X be a nonempty set. Define $d : X \times X \rightarrow \mathbb{R}$ by

$$d(x, y) = \begin{cases} 1, & x \neq y, \\ 0, & x = y. \end{cases}$$

It is immediate to verify (nonnegativity, identity, symmetry). For the triangle inequality: if $x \neq z$ then $d(x, z) = 1 \leq d(x, y) + d(y, z)$ since the right-hand side is 1, or 2 (cannot be 0 bc that implies $x = z$); if $x = z$, then $d(x, z) = 0 \leq d(x, y) + d(y, z)$. Therefore, d is a metric (the *discrete metric*).

Bounded Real Functions

Definition (Bounded functions). Fix $X \neq \emptyset$. A real function $f : X \rightarrow \mathbb{R}$ is **bounded** if $f(X) \subseteq \mathbb{R}$ is a bounded set. We denote

$$\mathcal{F}^B(X, \mathbb{R}) = \{ f : X \rightarrow \mathbb{R} : f \text{ is bounded} \}.$$

Remark. If f or g are not bounded, the set $\{|f(x) - g(x)| : x \in X\}$ may fail to have a finite upper bound (and its “supremum” would be $+\infty$), which would prevent defining a distance with values in $\mathbb{R}$. That is why we restrict the domain to $\mathcal{F}^B(X, \mathbb{R})$.

Supremum (Uniform) Metric on $\mathcal{F}^B(X, \mathbb{R})$

Definition (Supremum metric). Let $X \neq \emptyset$. Define $d_\infty : \mathcal{F}^B(X, \mathbb{R}) \times \mathcal{F}^B(X, \mathbb{R}) \rightarrow \mathbb{R}$ by

$$d_\infty(f, g) = \sup_{x \in X} |f(x) - g(x)|.$$

Note that $d_\infty(f, g) = \|f - g\|_\infty$, where $\|h\|_\infty := \sup_{x \in X} |h(x)|$.

Proposition. d_∞ is a metric on $\mathcal{F}^B(X, \mathbb{R})$.

Proof. Let $f, g, h \in \mathcal{F}^B(X, \mathbb{R})$.

- (a) **Nonnegativity:** $|f(x) - g(x)| \geq 0$ for all x , so the supremum is ≥ 0 .
- (b) **Identity of indiscernibles:** $d_\infty(f, g) = 0$ implies $\sup_x |f(x) - g(x)| = 0$, hence $|f(x) - g(x)| = 0$ for all x , i.e. $f = g$. The converse is clear.
- (c) **Symmetry:** $|f(x) - g(x)| = |g(x) - f(x)|$ for all x , so the suprema coincide.
- (d) **Triangle inequality:** by the usual triangle inequality in $\mathbb{R}$, $|f(x) - h(x)| \leq |f(x) - g(x)| + |g(x) - h(x)|$ for all $x \in X$. Taking suprema on both sides and using the fact

$$(\forall x, a_x \leq b_x) \implies \sup_x a_x \leq \sup_x b_x,$$

we obtain

$$\sup_x |f(x) - h(x)| \leq \sup_x |f(x) - g(x)| + \sup_x |g(x) - h(x)|.$$

That is, $d_\infty(f, h) \leq d_\infty(f, g) + d_\infty(g, h)$.

□

Lemma. If for every x one has $A(x) \leq B(x)$, then $\sup_x A(x) \leq \sup_x B(x)$. Reason: every supremum is the **least upper bound**; since $\sup_x B(x)$ bounds each $A(x)$, it also bounds their set of values and, by minimality, dominates the supremum of A .

Remark (When the supremum in d_∞ exists). If X is compact and f, g are continuous, then $x \mapsto |f(x) - g(x)|$ attains a maximum (extreme value theorem), and $d_\infty(f, g)$ exists without needing to assume that f, g are globally bounded in $\mathbb{R}$. In general, it suffices that $f - g$ be bounded.

Sequences: Boundedness, Monotonicity, and Basic Theorems

Definition (Bounded sequence in a metric space). Let (X, d) be a metric space and $(x_n)_{n \in \mathbb{N}}$ a sequence in X . We say that (x_n) is *bounded* if there exists some $y \in X$ and $M > 0$ such that

$$d(x_n, y) \leq M \quad \text{for all } n \in \mathbb{N}.$$

In particular, in $(\mathbb{R}, d_u)$ with $d_u(x, y) = |x - y|$, a real sequence (x_n) is bounded if and only if

$$\exists M > 0 \text{ such that } |x_n| \leq M \quad \forall n.$$

Indeed, if $|x_n - y| \leq M$ for all n (with some $y \in \mathbb{R}$), then

$$|x_n| \leq |x_n - y| + |y| \leq M + |y| \quad \forall n,$$

and therefore it is bounded taking $\widehat{M} := M + |y|$.

Remark (Boundedness of the set of values). For a real sequence (x_n) , the set of its values $E := \{x_n : n \geq 1\}$ is bounded if there exist $a, b \in \mathbb{R}$ with

$$a \leq x_n \leq b \quad \forall n.$$

Then $|x_n| \leq \max\{|a|, |b|\}$ for all n .

Proposition. Let (X, d) be a metric space. If (x_n) converges in X , then (x_n) is bounded.

Proof. Let $x \in X$ be the limit of (x_n) . Taking $\varepsilon = 1$, there exists N such that $d(x_n, x) \leq 1$ for all $n \geq N$. Define

$$M := \max\{1, d(x_1, x), \dots, d(x_{N-1}, x)\}.$$

Then $d(x_n, x) \leq M$ for all n , and by definition (x_n) is bounded. $\square$

Example. The sequence $((-1)^n)_{n \in \mathbb{N}} = (-1, 1, -1, 1, \dots)$ is bounded in $(\mathbb{R}, d_u)$ (since $|(-1)^n| \leq 1$) but does not converge.

Definition (Monotone sequences). A real sequence (x_n) is

- *increasing* if $x_{n+1} \geq x_n$ for all n ;
- *decreasing* if $x_{n+1} \leq x_n$ for all n ;
- *monotone* if it is either increasing or decreasing.

Proposition (Monotone convergence theorem). Let (x_n) be a monotone and bounded real sequence. Then (x_n) converges. Moreover:

$$\begin{aligned} \text{if } (x_n) \text{ is increasing,} \quad & \lim_{n \rightarrow \infty} x_n = \sup\{x_n : n \in \mathbb{N}\}; \\ \text{if } (x_n) \text{ is decreasing,} \quad & \lim_{n \rightarrow \infty} x_n = \inf\{x_n : n \in \mathbb{N}\}. \end{aligned}$$

Proof (increasing case). Let $S := \{x_n : n \in \mathbb{N}\}$; by boundedness, S has an upper bound, and by completeness of $\mathbb{R}$ there exists $x^* := \sup S$. Let $\varepsilon > 0$. By the definition of supremum, there exists N such that $x_N > x^* - \varepsilon$. Since (x_n) is increasing, for every $n \geq N$ we have $x_n \geq x_N > x^* - \varepsilon$ and, since x^* is an upper bound, $x_n \leq x^*$. Hence $0 \leq x^* - x_n < \varepsilon$, i.e. $|x_n - x^*| < \varepsilon$ for all $n \geq N$. Therefore $x_n \rightarrow x^*$. The decreasing case is analogous replacing sup by inf. $\square$

Remark. Proposition , together with the boundedness criterion of Proposition , is useful to *verify that a sequence does not converge*: if it is monotone but not bounded, it cannot converge.

Lemma (Monotone subsequence). Every sequence (x_n) in $(\mathbb{R}, d_u)$ has a monotone subsequence.

Corollary (Bolzano–Weierstrass). Every bounded sequence in $(\mathbb{R}, d_u)$ has a convergent subsequence.

Proposition (Squeeze theorem). Let $(x_n), (y_n), (z_n)$ be real sequences such that, for all n ,

$$x_n \leq y_n \leq z_n, \quad \lim_{n \rightarrow \infty} x_n = \lim_{n \rightarrow \infty} z_n = a \in \mathbb{R}.$$

Then $\lim_{n \rightarrow \infty} y_n = a$.

Proposition (Algebra of limits). Let (x_n) and (y_n) be real sequences with $\lim_{n \rightarrow \infty} x_n = x \in \mathbb{R}$ and $\lim_{n \rightarrow \infty} y_n = y \in \mathbb{R}$. Then:

1. $\lim_{n \rightarrow \infty} |x_n| = |x|$;
2. $\lim_{n \rightarrow \infty} (x_n + y_n) = x + y$;
3. $\lim_{n \rightarrow \infty} (x_n y_n) = x y$;
4. $\lim_{n \rightarrow \infty} (x_n / y_n) = x / y$ when $y \neq 0$ and $y_n \neq 0$.

Example (Example 4). Consider the sequence

$$x_n = \frac{3(1 - \frac{1}{n})^2}{2(n+1)} + \left(\lambda + \frac{1}{n}\right).$$

Since $(1 - \frac{1}{n})^2 \rightarrow 1$ and $\frac{2(n+1)}{n} \rightarrow 2$, the first term tends to $\frac{3}{2}$; moreover $\lambda + \frac{1}{n} \rightarrow \lambda$. Therefore

$$\lim_{n \rightarrow \infty} x_n = \lambda + \frac{3}{2}.$$

Application of Limits and Bounds: O and o Notation

Let (x_n) and (y_n) be two real sequences with $y_n \neq 0$ for all n .

Definition (Same order / Big- O). We say that (x_n) is of the *order* of (y_n) if the quotient $(\frac{x_n}{y_n})$ is bounded, that is, there exists $\Pi > 0$ such that

$$\left| \frac{x_n}{y_n} \right| \leq \Pi \quad \text{for all } n,$$

equivalently,

$$|x_n| \leq \Pi |y_n| \quad \forall n.$$

In this case we write $x_n = O(y_n)$. As a particular case, if $y_n \equiv 1$, then $x_n = O(1)$ means that (x_n) is bounded.

Definition (Lower order / little- o). We say that (x_n) is of *lower order* than (y_n) if

$$\lim_{n \rightarrow \infty} \frac{x_n}{y_n} = 0,$$

and we write $x_n = o(y_n)$. In particular, if $y_n \equiv 1$, then $x_n = o(1)$ is equivalent to $x_n \rightarrow 0$.

Proposition (Basic properties of O and o). Let $(x_n), (\hat{x}_n), (y_n), (\hat{y}_n)$ be real sequences.

1. If $x_n = O(y_n)$ and $\hat{x}_n = O(\hat{y}_n)$, then

$$x_n \hat{x}_n = O(y_n \hat{y}_n).$$

Proof: $|x_n| \leq \Pi |y_n|$ and $|\hat{x}_n| \leq \hat{\Pi} |\hat{y}_n| \Rightarrow |x_n \hat{x}_n| \leq (\Pi \hat{\Pi}) |y_n \hat{y}_n|$.

2. If $k \neq 0$ and $x_n = O(y_n)$, then $k x_n = O(|k| y_n)$.

3. If $x_n = O(y_n)$ and $\hat{x}_n = O(\hat{y}_n)$, then

$$x_n + \hat{x}_n = O(\max\{|y_n|, |\hat{y}_n|\}).$$

In particular, $|x_n + \hat{x}_n| \leq (\Pi + \hat{\Pi}) \max\{|y_n|, |\hat{y}_n|\} \leq 2 \max\{\Pi, \hat{\Pi}\} \max\{|y_n|, |\hat{y}_n|\}.$

4. If $x_n = O(y_n)$ and also $\hat{x}_n = O(y_n)$, then $x_n + \hat{x}_n = O(y_n)$. (This is a particular case of (3).)

5. If $k \neq 0$ and $x_n = o(y_n)$, then $kx_n = o(y_n)$.

6. If $x_n = o(y_n)$ and $\hat{x}_n = o(\hat{y}_n)$, then

$$x_n \hat{x}_n = o(y_n \hat{y}_n),$$

$$\text{since } \frac{x_n \hat{x}_n}{y_n \hat{y}_n} = \frac{x_n}{y_n} \cdot \frac{\hat{x}_n}{\hat{y}_n} \rightarrow 0 \cdot 0 = 0.$$

7. If $x_n = o(y_n)$ and $\hat{x}_n = o(\hat{y}_n)$, then $x_n + \hat{x}_n = o(y_n)$, because $\frac{x_n + \hat{x}_n}{y_n} = \frac{x_n}{y_n} + \frac{\hat{x}_n}{y_n} \rightarrow 0 + 0 = 0.$

Cauchy and a Clarification on Boundedness

Initial clarification

Given a metric space (X, d) and a sequence $(x_n)_{n \in \mathbb{N}} \subset X$:

- $(x_1, x_2, x_3, \dots)$ is a *sequence*; in contrast, $\{x_1, x_2, x_3, \dots\}$ is the *set* of values the sequence may produce.
- We say that (x_n) is *bounded* if there exists $y \in X$ and $M > 0$ such that $d(x_n, y) \leq M$ for all $n \in \mathbb{N}$. For a fixed $y \in X$, let

$$E_y = \{d(x_n, y) : n \in \mathbb{N}\} \subset \mathbb{R}.$$

Then (x_n) is bounded $\iff E_y$ is bounded above. Moreover, if $d(x_n, y) \leq M$ for all n and $y' \in X$ is arbitrary, by the triangle inequality

$$d(x_n, y') \leq d(x_n, y) + d(y, y') \leq M + d(y, y'),$$

so the notion of boundedness does not depend on the chosen center.

- Trivial case: if the set of values $\{x_n : n \in \mathbb{N}\}$ is finite, then (x_n) is bounded. The case that requires attention is when *infinitely many* distinct values appear.
- **Observation.** An unbounded sequence does *not necessarily* contain a bounded subsequence (e.g. $x_n = n$ in $\mathbb{R}$ has none).

Remark. We will say that a subsequence is *proper* when it does not coincide with the original sequence.

Cauchy Sequences

Definition (Cauchy sequence). Let (X, d) be a metric space and $(x_n)_{n \in \mathbb{N}} \subset X$. We say that (x_n) is *Cauchy* if for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that for all $m, n \geq N$ we have

$$d(x_n, x_m) < \varepsilon.$$

Example (1b). In $(\mathbb{R}, d_u)$ (the usual metric), the sequence $x_n = \frac{1}{n}$ is Cauchy. Indeed, given $\varepsilon > 0$, choose N with $\frac{1}{N} < \varepsilon$. If $m, n \geq N$, then

$$|x_n - x_m| = \left| \frac{1}{n} - \frac{1}{m} \right| \leq \frac{1}{N} < \varepsilon.$$

(Note that in $(0, 1)$ the same sequence does *not* converge because its limit $0 \notin (0, 1)$.)

Example (3). The sequence $x_n = (-1)^n$ in $(\mathbb{R}, d_u)$ is not Cauchy. Take $\varepsilon \in (0, 1)$. For every N we have

$$|x_{N+1} - x_N| = |(-1)^{N+1} - (-1)^N| = 2 > \varepsilon,$$

hence the Cauchy condition fails.

Proposition (7). *Every convergent sequence is Cauchy.*

Proof. Let (x_n) be a sequence in (X, d) that converges to $x \in X$. Given $\varepsilon > 0$, by convergence there exists N such that $d(x_n, x) < \varepsilon/2$ for all $n \geq N$. If $m, n \geq N$, by the triangle inequality,

$$d(x_n, x_m) \leq d(x_n, x) + d(x, x_m) < \varepsilon/2 + \varepsilon/2 = \varepsilon.$$

□

Remark. The converse does not always hold in an arbitrary metric space: a sequence may be Cauchy and not converge if the space is not complete (e.g. $1/n$ in $(0, 1)$).

Proposition (Convergent $\Rightarrow$ Cauchy). *Let $(x_n)_{n \in \mathbb{N}}$ be a sequence in a metric space (X, d) that converges to $x \in X$. Then (x_n) is Cauchy.*

Proof. Fix $\varepsilon > 0$. By convergence, there exists N such that $d(x_n, x) < \varepsilon/2$ for all $n \geq N$. If $m, n \geq N$, then by the triangle inequality,

$$d(x_n, x_m) \leq d(x_n, x) + d(x, x_m) < \varepsilon/2 + \varepsilon/2 = \varepsilon.$$

□

Proposition (Every Cauchy sequence is bounded). *Let $(x_n)_{n \in \mathbb{N}}$ be a Cauchy sequence in (X, d) . Then there exist $y \in X$ and $M > 0$ such that $d(x_n, y) \leq M$ for all n (that is, (x_n) is bounded).*

Proof. Take $\varepsilon = 1$. Since (x_n) is Cauchy, there exists N such that $d(x_n, x_m) \leq 1$ for all $m, n \geq N$. Fix $y := x_N$. Then $d(x_n, y) \leq 1$ for all $n \geq N$. For the finitely many indices $1 \leq n < N$, define

$$K := \max\{d(x_1, y), d(x_2, y), \dots, d(x_{N-1}, y)\}.$$

Let $M := \max\{K, 1\}$. Then $d(x_n, y) \leq M$ for all $n \in \mathbb{N}$.

□

Definition (Complete space). We say that a metric space (X, d) is *complete* if every Cauchy sequence in (X, d) converges (to a point in X).

Example.

- $(\mathbb{R}^n, d_{\text{eucl}})$ is complete.
- Every finite set with the discrete metric is complete.
- If $\mathcal{F}^B(X, \mathbb{R})$ denotes the space of *bounded* functions $f : X \rightarrow \mathbb{R}$ and $d_\infty(f, g) := \sup_{x \in X} |f(x) - g(x)|$, then $(\mathcal{F}^B(X, \mathbb{R}), d_\infty)$ is complete.

Example (Incomplete spaces).

- $((0, 1), d_u)$ is not complete (e.g. $x_n = \frac{1}{n}$ is Cauchy but does not converge in $(0, 1)$).
- $(\mathbb{Q}, d_u)$ is not complete (there are Cauchy sequences that “aim” at irrational numbers. For example, any sequence of rational truncations of $\sqrt{2}$ is Cauchy in $(\mathbb{Q}, d_u)$ but has no limit in $\mathbb{Q}$).

Divergent Sequences and Cluster Points

Example. In $(\mathbb{R}, d_u)$ consider

$$x_n = (-1)^n \left(1 - \frac{1}{n}\right) = \begin{cases} 1 - \frac{1}{n}, & n \text{ even}, \\ -1 + \frac{1}{n}, & n \text{ odd}. \end{cases}$$

The sequence does not converge. Indeed, its even and odd subsequences satisfy

$$x_{2m} = 1 - \frac{1}{2m} \longrightarrow 1, \quad x_{2m-1} = -1 + \frac{1}{2m-1} \longrightarrow -1.$$

Since a convergent sequence cannot have two subsequences converging to different limits, (x_n) is divergent. Nevertheless, it approaches both 1 and -1 infinitely often.

Definition (Cluster point of a sequence). Let $(x_n)_{n \in \mathbb{N}}$ be a sequence in a metric space (X, d) . A point $x \in X$ is a *cluster point* of (x_n) if

$$\forall \varepsilon > 0, \forall N \in \mathbb{N}, \exists n \geq N \text{ such that } d(x_n, x) < \varepsilon.$$

Thus, every neighborhood of x contains terms of the sequence with arbitrarily large indices.

Proposition (Cluster points and subsequences). *Let (x_n) be a sequence in a metric space (X, d) . A point $x \in X$ is a cluster point of (x_n) if and only if there exists a subsequence (x_{n_k}) such that*

$$x_{n_k} \longrightarrow x.$$

Proof. Suppose first that $x_{n_k} \rightarrow x$. Given $\varepsilon > 0$ and $N \in \mathbb{N}$, choose k sufficiently large that

$$d(x_{n_k}, x) < \varepsilon \quad \text{and} \quad n_k \geq N.$$

Hence x is a cluster point.

Conversely, suppose that x is a cluster point. We construct a subsequence recursively. Since x is a cluster point, there exists $n_1 \geq 1$ such that

$$d(x_{n_1}, x) < 1.$$

Having chosen n_k , use the cluster-point property with $\varepsilon = 1/(k+1)$ and $N = n_k + 1$ to choose $n_{k+1} > n_k$ satisfying

$$d(x_{n_{k+1}}, x) < \frac{1}{k+1}.$$

Therefore

$$d(x_{n_k}, x) < \frac{1}{k} \longrightarrow 0,$$

and hence $x_{n_k} \rightarrow x$. □

Proposition (Cluster points of Example). *The sequence*

$$x_n = (-1)^n \left(1 - \frac{1}{n}\right)$$

has exactly two cluster points:

$$\{-1, 1\}.$$

Proof. The even subsequence satisfies

$$x_{2m} = 1 - \frac{1}{2m} \longrightarrow 1,$$

so 1 is a cluster point. Similarly,

$$x_{2m-1} = -1 + \frac{1}{2m-1} \longrightarrow -1,$$

so -1 is a cluster point.

It remains to show that there are no others. Let (x_{n_k}) be any convergent subsequence, and suppose

$$x_{n_k} \longrightarrow L.$$

Among the indices (n_k) , either there are infinitely many even indices or infinitely many odd indices.

If there are infinitely many even indices, we can extract from (x_{n_k}) a further subsequence consisting entirely of even-indexed terms. Since every even-indexed term belongs to the sequence

$$x_{2m} = 1 - \frac{1}{2m},$$

this further subsequence converges to 1. But every subsequence of the convergent sequence (x_{n_k}) must converge to the same limit L . Therefore $L = 1$.

Likewise, if there are infinitely many odd indices, we can extract a further subsequence consisting entirely of odd-indexed terms. Since

$$x_{2m-1} = -1 + \frac{1}{2m-1} \longrightarrow -1,$$

we obtain $L = -1$.

Therefore every convergent subsequence has limit either 1 or -1 . By the characterization of cluster points through convergent subsequences, these are the only cluster points. $\square$

Cluster Points, $\limsup/\liminf$, and Open Sets

Some Basic Facts about Cluster Points

Proposition. *Let (x_n) be a sequence in a metric space (X, d) .*

1. *If (x_n) converges to x , then x is its unique cluster point.*
2. *An unbounded sequence may nevertheless have cluster points. For example,*

$$x_n = \begin{cases} 1, & n \text{ odd}, \\ n, & n \text{ even}, \end{cases} \quad \text{in } (\mathbb{R}, d_u),$$

is unbounded, but 1 is a cluster point.

Proof. For part (1), if $x_n \rightarrow x$, then the sequence itself is a subsequence converging to x , so x is a cluster point.

To prove uniqueness, suppose that $y \neq x$. Let

$$\varepsilon := \frac{d(x, y)}{3} > 0.$$

Since $x_n \rightarrow x$, there exists N such that

$$d(x_n, x) < \varepsilon \quad \text{for all } n \geq N.$$

By the triangle inequality,

$$d(x, y) \leq d(x, x_n) + d(x_n, y),$$

so for $n \geq N$,

$$d(x_n, y) \geq d(x, y) - d(x_n, x) > 3\varepsilon - \varepsilon = 2\varepsilon.$$

Hence no term with sufficiently large index belongs to $B(y, \varepsilon)$, so y cannot be a cluster point.

For part (2), the odd subsequence is constant:

$$x_{2k-1} = 1 \quad \text{for every } k,$$

and therefore converges to 1. Hence 1 is a cluster point. $\square$

Theorem 1 (Bolzano–Weierstrass in $\mathbb{R}$). *Every bounded sequence in $(\mathbb{R}, d_u)$ has a convergent subsequence. Equivalently, every bounded real sequence has at least one cluster point.*

Remark (Caution with the ambient space). The ambient metric space matters. In the subspace

$$((0, 1], d_u),$$

the sequence

$$x_n = \frac{1}{n}$$

is bounded but has no cluster point in $(0, 1]$.

Indeed, every subsequence of $(1/n)$ converges in $\mathbb{R}$ to 0, but

$$0 \notin (0, 1].$$

Thus Bolzano–Weierstrass applies to bounded sequences in $\mathbb{R}$, but it does not imply that every bounded sequence in an arbitrary metric space has a cluster point belonging to that space.

lim sup and lim inf

Definition (Extended real line). The *extended real line* is

$$\overline{\mathbb{R}} := \mathbb{R} \cup \{-\infty, +\infty\}.$$

We use the conventions that if a set is not bounded above its supremum is $+\infty$, and if it is not bounded below its infimum is $-\infty$.

Definition (lim sup and lim inf). Let (x_n) be a real sequence. For each n , define the supremum and infimum of the n th tail by

$$s_n := \sup_{k \geq n} x_k, \quad i_n := \inf_{k \geq n} x_k,$$

where these values are allowed to lie in $\overline{\mathbb{R}}$.

The *limit superior* and *limit inferior* are defined by

$$\limsup_{n \rightarrow \infty} x_n := \inf_{n \in \mathbb{N}} s_n = \inf_{n \in \mathbb{N}} \sup_{k \geq n} x_k,$$

and

$$\liminf_{n \rightarrow \infty} x_n := \sup_{n \in \mathbb{N}} i_n = \sup_{n \in \mathbb{N}} \inf_{k \geq n} x_k.$$

Thus

$$\limsup_{n \rightarrow \infty} x_n, \quad \liminf_{n \rightarrow \infty} x_n \in \overline{\mathbb{R}}.$$

Remark (Intuition from the tails). As n increases, the tail

$$\{x_k : k \geq n\}$$

becomes smaller. Therefore,

$$s_{n+1} \leq s_n, \quad i_{n+1} \geq i_n.$$

Hence (s_n) is decreasing and (i_n) is increasing. Informally, $\limsup x_n$ records the eventual upper behavior of the sequence, while $\liminf x_n$ records its eventual lower behavior.

Proposition (Basic properties of lim sup and lim inf). *Let (x_n) be a real sequence. Then:*

1.

$$\liminf_{n \rightarrow \infty} x_n \leq \limsup_{n \rightarrow \infty} x_n.$$

2. If

$$\limsup_{n \rightarrow \infty} x_n = L \in \mathbb{R},$$

then L is a cluster point of (x_n) .

3. If

$$\liminf_{n \rightarrow \infty} x_n = \ell \in \mathbb{R},$$

then ℓ is a cluster point of (x_n) .

4. If (x_n) is bounded, then both $\limsup x_n$ and $\liminf x_n$ are finite. Moreover,

$$\limsup_{n \rightarrow \infty} x_n = \max\{x \in \mathbb{R} : x \text{ is a cluster point of } (x_n)\},$$

and

$$\liminf_{n \rightarrow \infty} x_n = \min\{x \in \mathbb{R} : x \text{ is a cluster point of } (x_n)\}.$$

5. For any $x \in \mathbb{R}$,

$$x_n \rightarrow x \iff \liminf_{n \rightarrow \infty} x_n = \limsup_{n \rightarrow \infty} x_n = x.$$

Remark. The boundedness qualification in part (4) is important. For example, consider

$$x_n = \begin{cases} 1, & n \text{ odd}, \\ n, & n \text{ even}. \end{cases}$$

Then

$$\liminf_{n \rightarrow \infty} x_n = 1, \quad \limsup_{n \rightarrow \infty} x_n = +\infty.$$

The point 1 is a cluster point in $\mathbb{R}$, whereas $+\infty$ is not a point of $\mathbb{R}$ and therefore is not a cluster point in the metric space $(\mathbb{R}, d_u)$.

Example. Let

$$x_n = (-1)^n \left(1 - \frac{1}{n}\right).$$

For every fixed n , the tail $\{x_k : k \geq n\}$ contains even-indexed terms arbitrarily close to 1 from below. Therefore,

$$\sup_{k \geq n} x_k = 1 \quad \text{for every } n.$$

Notice that the supremum need not be attained by any term of the tail.

Similarly, every tail contains odd-indexed terms arbitrarily close to -1 from above, so

$$\inf_{k \geq n} x_k = -1 \quad \text{for every } n.$$

Hence

$$\limsup_{n \rightarrow \infty} x_n = \inf_{n \in \mathbb{N}} 1 = 1,$$

and

$$\liminf_{n \rightarrow \infty} x_n = \sup_{n \in \mathbb{N}} (-1) = -1.$$

This agrees with the cluster-point calculation above:

$$\text{Clust}(x_n) = \{-1, 1\},$$

with 1 the largest cluster point and -1 the smallest.

Open Balls, Interior Points, and Open Sets

Definition (Open ball). In a metric space (X, d) and for $x \in X$, $\varepsilon > 0$, the *open ball* centered at x with radius ε is

$$B(x, \varepsilon) = \{y \in X : d(x, y) < \varepsilon\}.$$

Example. 1. In $(\mathbb{R}, d_u)$, $B(x, \varepsilon) = (x - \varepsilon, x + \varepsilon)$.

2. In the subspace $([0, 1], d_u)$,

$$B(1/2, 3/4) = (-\frac{1}{4}, \frac{5}{4}) \cap [0, 1] = [0, 1].$$

3. In $(\mathbb{R}^2, \text{Euclidean distance})$, $B((a, b), \varepsilon)$ is the usual open disk.

Definition (Interior point and interior). Let $E \subset X$. A point $x \in E$ is *interior* to E if there exists $\varepsilon > 0$ such that $B(x, \varepsilon) \subset E$. The *interior* of E is

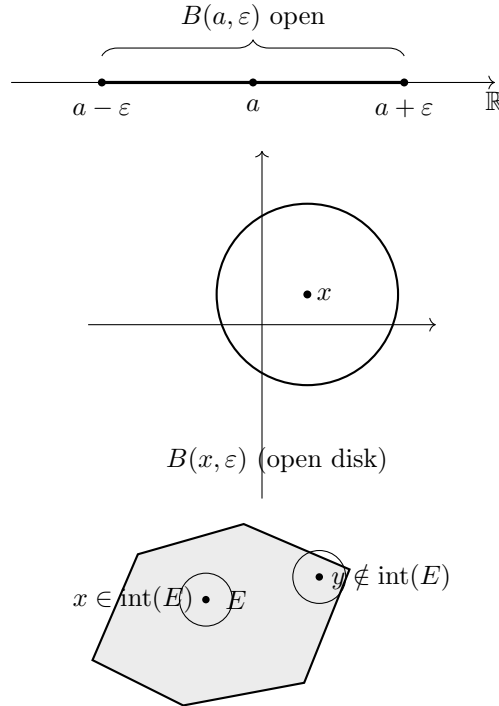
$$\text{int}(E) = \{x \in E : x \text{ is interior to } E\}.$$

Definition (Open set). A set $E \subset X$ is *open* if $E = \text{int}(E)$, i.e., if every point of E is interior.

Remark. 1. For every E , always $\text{int}(E) \subset E$.

2. E is open $\iff E \subset \text{int}(E)$.

3. Openness depends on the ambient space: in $(\mathbb{R}, d_u)$ the set $[0, 1]$ is not open (because of 0), whereas in the subspace $([0, 1], d_u)$ the set $(0, 1]$ is relatively open.



Cluster points and subsequences

Definition (Cluster point). Let $(x_n)_{n \in \mathbb{N}}$ be a sequence in a metric space (X, d) . We say that $x \in X$ is a *cluster point* of (x_n) if for every $\varepsilon > 0$ and every $N \in \mathbb{N}$ there exists $n \geq N$ such that $d(x_n, x) < \varepsilon$.

- If a *subsequence* (x_{n_k}) converges to x^* , then x^* is a cluster point of the original sequence (x_n) .

- If (x_n) converges to x , then x is a cluster point of (x_n) .
- An unbounded sequence can have a cluster point. For example, $x_{2k} = k$ and $x_{2k+1} = 1$ is unbounded and has 1 as a cluster point.
- In $(\mathbb{R}, d_\mu)$ (the usual metric), every bounded sequence has a cluster point (Bolzano–Weierstrass): it has a convergent subsequence and its limit is a cluster point.
- In the subspace $((0, 1], d_\mu)$, the sequence $x_n = \frac{1}{n}$ is bounded but does *not* have a cluster point in X (its only natural limit would be $0 \notin X$).

Upper and lower limits

Definition (lim sup and lim inf). For a real sequence (x_n) define

$$\limsup_{n \rightarrow \infty} x_n := \inf_{n \in \mathbb{N}} \sup_{k \geq n} x_k, \quad \liminf_{n \rightarrow \infty} x_n := \sup_{n \in \mathbb{N}} \inf_{k \geq n} x_k.$$

The lim sup (resp. lim inf) is the largest (resp. smallest) cluster point if (x_n) is bounded.

Example. For $x_n = (-1)^n(1 - \frac{1}{n})$ we have

$$\limsup_{n \rightarrow \infty} x_n = 1, \quad \liminf_{n \rightarrow \infty} x_n = -1.$$

Indeed, fixing n , the suprema (resp. infima) of the tails $\{x_k : k \geq n\}$ approach 1 (resp. -1).

Open balls, interior points, and open sets

Definition (Open ball). Given (X, d) , $x \in X$, and $\varepsilon > 0$, the *open ball* centered at x with radius ε is

$$B(x, \varepsilon) = \{y \in X : d(x, y) < \varepsilon\}.$$

Example.

1. In $(\mathbb{R}, d_\mu)$: $B(x, \varepsilon) = (x - \varepsilon, x + \varepsilon)$.
2. In the subspace $([0, 10], d_\mu)$:

$$B(\frac{1}{2}, \frac{3}{4}) = (-\frac{1}{4}, \frac{5}{4}) \cap [0, 10] = [0, \frac{5}{4}).$$

3. In $(\mathbb{R}^2, d_{\text{euc}})$: $B(x, \varepsilon)$ is the open disk of radius ε .

Definition (Interior point and interior). Let $E \subset X$. A point $x \in E$ is *interior* to E if there exists $\varepsilon > 0$ such that $B(x, \varepsilon) \subset E$. The *interior* of E is $\text{int}(E) = \{x \in E : x \text{ is interior}\}$.

Definition (Open set). A set $E \subset X$ is *open* if $E = \text{int}(E)$, equivalently, if every $x \in E$ is an interior point.

Remark. Openness depends on the underlying space. For example, in $(\mathbb{R}, d_\mu)$, $[0, 1]$ is not open; however, in the subspace $((0, 100), d_\mu)$ the set $[0, 1]$ is open.

Example.

1. In the discrete metric (X, d_{disc}) , $\{x\}$ is open because $B(x, \frac{1}{2}) = \{x\}$. In fact, *every* subset of X is open.
2. In $(\mathbb{R}, d_\mu)$, no finite subset (nor, more generally, any countable set) is open: every open ball in $\mathbb{R}$ contains infinitely many (indeed uncountably many) points. Exception: the $\emptyset$ set (finite, countable and open).

Basic properties of open sets

Proposition. In a metric space (X, d) :

1. $\emptyset$ and X are open.
2. Arbitrary unions of open sets are open.
3. Finite intersections of open sets are open.

Example. A countable intersection of open sets need not be open in $(\mathbb{R}, d_\mu)$:

$$\bigcap_{n \geq 1} \left(-\frac{1}{n}, \frac{1}{n}\right) = \{0\},$$

which is not open.

Proposition. For every $x \in X$ and $\varepsilon > 0$, the open ball $B(x, \varepsilon)$ is an open set. In particular, $\text{int}(E)$ is always an open set and $\text{int}(E) \subset E$.

Closed balls and closed sets

Definition (Closed ball and closed set). The *closed ball* centered at x with radius $\varepsilon > 0$ is

$$\overline{B}(x, \varepsilon) = \{y \in X : d(x, y) \leq \varepsilon\}.$$

A set $F \subset X$ is *closed* if $X \setminus F$ is open.

Proposition. Every closed ball $\overline{B}(x, \varepsilon)$ is a closed set.

Remark (The real line $\mathbb{R}$). In $(\mathbb{R}, d_\mu)$, the complement of a closed interval $[x - \varepsilon, x + \varepsilon]$ is open: if $y \notin [x - \varepsilon, x + \varepsilon]$, then taking $\varepsilon' = \frac{1}{2} \text{dist}(y, [x - \varepsilon, x + \varepsilon]) > 0$ we have $B(y, \varepsilon') \subset \mathbb{R} \setminus [x - \varepsilon, x + \varepsilon]$.

Lemma. Let (x_n) be a sequence in a metric space (X, d) . A point $x^* \in X$ is a cluster point of (x_n) if and only if there exists a subsequence (x_{n_k}) such that $x_{n_k} \rightarrow x^*$.

Example. Consider in $(\mathbb{R}, d_u)$ the sequence

$$x_1 = 1, x_2 = 2, x_3 = 1, x_4 = 3, x_5 = 1, x_6 = 4, \dots$$

For each n we have $\sup\{x_k : k \geq n\} = +\infty$, so

$$\limsup_{n \rightarrow \infty} x_n = \inf_n \sup_{k \geq n} x_k = +\infty,$$

which is *not* a cluster point in $\mathbb{R}$. On the other hand, $\inf\{x_k : k \geq n\} = 1$ for all n , and thus

$$\liminf_{n \rightarrow \infty} x_n = \sup_n \inf_{k \geq n} x_k = 1,$$

and 1 is indeed a cluster point (taking the subsequence of odd terms).

Proposition. For any real sequence (x_n) :

1. If x^* is a cluster point, then $\liminf_{n \rightarrow \infty} x_n \leq x^* \leq \limsup_{n \rightarrow \infty} x_n$.
2. If $\limsup_{n \rightarrow \infty} x_n \in \mathbb{R}$, then this value is a cluster point (analogously for the $\liminf$).
3. The limit $\lim_{n \rightarrow \infty} x_n = x \in \mathbb{R}$ exists if and only if $\liminf_{n \rightarrow \infty} x_n = \limsup_{n \rightarrow \infty} x_n = x$.

Closed sets

Proposition. In a metric space (X, d) :

1. $\emptyset$ and X are closed.
2. The finite union of closed sets is closed.
3. The arbitrary intersection of closed sets is closed.

Example. In $(\mathbb{R}, d_u)$, for $n \in \mathbb{N}$ let $F_n = [-1 + 1/n, 1 - 1/n]$. Each F_n is closed, but

$$\bigcup_{n \in \mathbb{N}} F_n = (-1, 1),$$

which is not closed (its complement is not open; at $x = 1$ there is no $\varepsilon > 0$ such that $B(x, \varepsilon) \subseteq (-\infty, -1] \cup [1, \infty)$).

Proposition (Sequential characterization of closed sets). *A set $E \subseteq X$ is closed if and only if for every convergent sequence (x_n) with $x_n \in E$ for all n , we have $\lim_{n \rightarrow \infty} x_n \in E$.*

Idea. If E is closed and $x_n \rightarrow x$ with $x_n \in E$, assuming $x \notin E$ implies $x \in X \setminus E$, which is open; hence there exists $\varepsilon > 0$ with $B(x, \varepsilon) \subseteq X \setminus E$. But then, for n large, $x_n \in B(x, \varepsilon) \subseteq X \setminus E$, a contradiction. The converse follows by considering complements. $\square$

Proposition (Sequential characterization of closed sets). *Let (X, d) be a metric space and $E \subseteq X$. The following are equivalent:*

1. E is closed.
2. For every sequence $\{x_n\}_{n \in \mathbb{N}}$ with $x_n \in E$ and $x_n \xrightarrow{n \rightarrow \infty} x \in X$, it follows that $x \in E$.

Proof. (1) $\Rightarrow$ (2): If E is closed, then $X \setminus E$ is open. If there were $x \in X \setminus E$ and $x_n \in E$ with $x_n \rightarrow x$, choose $\varepsilon > 0$ such that $B(x, \varepsilon) \subseteq X \setminus E$, contradicting the fact that eventually $x_n \in B(x, \varepsilon)$.

(2) $\Rightarrow$ (1) (contrapositive): Suppose E is not closed. Then $X \setminus E$ is not open: there exists $x \in X \setminus E$ that is not an interior point of $X \setminus E$. Hence, for every $n \in \mathbb{N}$,

$$B(x, 1/n) \cap E \neq \emptyset.$$

Choose $x_n \in B(x, 1/n) \cap E$; then $x_n \rightarrow x$ and $x \notin E$. Thus we obtain a sequence of points in E converging to a point outside E , contradicting (2). Therefore, E is closed. $\square$

Remark (Practical guide). To show that E is closed, it suffices to:

1. Prove that $X \setminus E$ is open; **or**
2. Prove that every convergent sequence in E has its limit in E .

To show that E is *not* closed, it suffices to:

1. Exhibit that $X \setminus E$ is not open; **or**
2. Find a sequence $\{x_n\}_{n \in \mathbb{N}}$ with $x_n \in E$ and $x_n \rightarrow x \notin E$.

Proposition (Operations with closed sets). *Let (X, d) be a metric space. Then:*

1. $\emptyset$ and X are closed.
2. A finite union of closed sets is closed.
3. An (arbitrary) intersection of closed sets is closed.

Example (Countable union of closed sets that is not closed). In $(\mathbb{R}, d_u)$ with the usual metric, for $n \in \mathbb{N}$ define

$$F_n = \left[-1 + \frac{1}{n}, 1 - \frac{1}{n}\right].$$

Each F_n is closed. However

$$\bigcup_{n \in \mathbb{N}} F_n = (-1, 1),$$

which is not closed (for instance, 1 is an adherent point but $1 \notin (-1, 1)$).

Interior, adherence, and closure

Definition (Interior point and interior). Let $E \subseteq X$. We say that $x \in E$ is an *interior point* of E if there exists $\varepsilon > 0$ such that $B(x, \varepsilon) \subseteq E$. The set of all interior points is denoted

$$\text{int}(\cdot)E = \{x \in E : \exists \varepsilon > 0 \text{ with } B(x, \varepsilon) \subseteq E\}$$

and is called the *interior* of E .

Proposition (Basic properties of the interior). *For every $E \subseteq X$:*

1. $\text{int}(\cdot)E$ is open and $\text{int}(\cdot)E \subseteq E$.
2. If U is open and $U \subseteq E$, then $U \subseteq \text{int}(\cdot)E$ (largest open set contained in E).
3. $\text{int}(\cdot)\text{int}(\cdot)E = \text{int}(\cdot)E$.

Definition (Adherent point and closure). Let $E \subseteq X$. We say that $x \in X$ is an *adherent point* of E if

$$\forall \varepsilon > 0, \quad B(x, \varepsilon) \cap E \neq \emptyset.$$

The *closure* of E is the set of its adherent points:

$$\overline{E} = \{x \in X : \forall \varepsilon > 0, B(x, \varepsilon) \cap E \neq \emptyset\}.$$

Proposition (Basic properties of the closure). *For every $E \subseteq X$:*

1. $\overline{E}$ is closed and $E \subseteq \overline{E}$.
2. If F is closed and $E \subseteq F$, then $\overline{E} \subseteq F$ (smallest closed set containing E).
3. We have the open-closed sandwich:

$$\text{int}(\cdot)E \subseteq E \subseteq \overline{E}.$$

Example (Set neither open nor closed in $\mathbb{R}$). In $(\mathbb{R}, d_u)$, the set $E = [0, 1)$ is not open (no ball around 0 fits inside E) and not closed (its complement $\mathbb{R} \setminus E = (-\infty, 0) \cup [1, \infty)$ is not open).

Example (Open ball plus isolated points). In $(\mathbb{R}^2, d_{\text{eucl}})$ fix $x \in \mathbb{R}^2$ and $\varepsilon > 0$. Let

$$F = B(x, \varepsilon) \cup \{p_1, \dots, p_m\},$$

where $p_i \in \partial B(x, \varepsilon)$ are isolated boundary points. Then:

- F is not open (no p_i has a ball $B(p_i, \delta)$ contained in F).
- F is not closed: for instance, a point q of the circle $\partial B(x, \varepsilon)$ different from the p_i is an adherent point of F but $q \notin F$. In particular,

$$\text{int}(\cdot)F = B(x, \varepsilon) \quad \text{and} \quad \overline{F} = \overline{B(x, \varepsilon)}.$$

Remark (Closure of open balls). In normed spaces (in particular in $\mathbb{R}^n$ with the Euclidean norm), $\overline{B(x, \varepsilon)} = \overline{B(x, \varepsilon)}$.

Definition (Adherent point, closure, and boundary). Let (X, d) be a metric space and $E \subseteq X$. We say that $x \in X$ is an *adherent point* of E if for every $\varepsilon > 0$ we have

$$B(x, \varepsilon) \cap E \neq \emptyset.$$

The *closure* of E is

$$\overline{E} = \text{cl}(E) = \{x \in X : x \text{ is an adherent point of } E\}.$$

The *boundary* of E is

$$\partial E = \overline{E} \setminus \text{int}(E).$$

Proposition. Let $E \subseteq X$. Then:

1. $\text{int}(E)$ is the largest open set contained in E .
2. $\overline{E}$ is the smallest closed set containing E .

Proof. (1) By definition, $x \in \text{int}(E)$ if there exists $\varepsilon > 0$ with $B(x, \varepsilon) \subseteq E$, hence $\text{int}(E)$ is open and $\text{int}(E) \subseteq E$. If U is open and $U \subseteq E$, then for each $x \in U$ we have $B(x, \varepsilon) \subseteq U \subseteq E$, so $x \in \text{int}(E)$ and $U \subseteq \text{int}(E)$.

(2) If $x \notin \overline{E}$, there exists $\varepsilon > 0$ with $B(x, \varepsilon) \cap E = \emptyset$, hence $B(x, \varepsilon) \subseteq X \setminus \overline{E}$, so $X \setminus \overline{E}$ is open and $\overline{E}$ is closed. If F is closed and $E \subseteq F$, every adherent point of E is also an adherent point of F , thus $\overline{E} \subseteq F$. $\square$

Definition (Cover and open cover). Let $E \subseteq X$. A family $\{F_i\}_{i \in I}$ of subsets of X is a *cover* of E if $E \subseteq \bigcup_{i \in I} F_i$. It is an *open cover* if, moreover, each F_i is open in X .

Definition (Compact set). We say that $E \subseteq X$ is *compact* if every open cover of E admits a finite subcover; that is, if $\{F_i\}_{i \in I}$ is an open cover of E , then there exists a finite set $J \subseteq I$ such that $E \subseteq \bigcup_{i \in J} F_i$. A metric space (X, d) is *compact* if X is compact as a subset of itself.

Example $((0, 1)$ is not compact in $(\mathbb{R}, d_{\text{us}}))$. Consider the family of open sets

$$\mathcal{U} = \{(1/n, 1) : n \geq 2\}.$$

We have $\bigcup_{n \geq 2} (1/n, 1) = (0, 1)$, so $\mathcal{U}$ is an open cover of $(0, 1)$. If we take a finite subfamily $\{(1/n_k, 1)\}_{k=1}^m$, then

$$\bigcup_{k=1}^m (1/n_k, 1) = (1/\max\{n_k\}_{k=1}^m, 1) \neq (0, 1),$$

since, for example, $x = \frac{1}{\max\{n_k\}_{k=1}^m + 1} \in (0, 1)$ is not covered. Therefore, no finite subcover exists and $(0, 1)$ is not compact.

Remark. Being compact does not mean “having some” finite open cover, but that *every* open cover admits a finite subcover.

Example $((0, 1)$ is not compact in $(\mathbb{R}, d_{\text{euc}}))$. Let $E = (0, 1)$ and consider the open cover

$$\mathcal{U} = \{U_n = (\frac{1}{n}, 1) : n \geq 2\}.$$

We have $\bigcup_{n \geq 2} U_n = (0, 1)$. If we take a finite subcover $\{U_{n_1}, \dots, U_{n_k}\}$, then $\bigcup_{j=1}^k U_{n_j} = (\frac{1}{N}, 1)$ with $N = \max\{n_1, \dots, n_k\}$, and thus points near 0 are not covered (for instance $x \in (0, \frac{1}{N})$). Therefore E does not admit a finite subcover and is not compact.

Example (A closed unbounded set that is not compact). In $(\mathbb{R}, d_{\text{euc}})$, let $E = [0, \infty)$. For $n \in \mathbb{Z}$ with $n \geq -1$, define $U_n = (n, n+2)$. Then $\bigcup_{n \geq -1} U_n = (-1, \infty) \supset E$, i.e. this is an open cover of E . No finite subcover can cover E : if we choose finitely many $U_{n_1}, \dots, U_{n_k}$ and $N = \max\{n_1, \dots, n_k\}$, then $\bigcup_{j=1}^k U_{n_j} \subset (-\infty, N+2)$, so $E \cap (N+3, \infty)$ remains uncovered. Hence E is not compact.

Definition (Bounded set). Let (X, d) be a metric space. We say that $E \subset X$ is *bounded* if there exist $x \in X$ and $\Pi > 0$ such that $d(x, y) \leq \Pi$ for all $y \in E$.

Theorem 2 (Sequential compactness in metric spaces). *A metric space is compact if and only if every sequence has a convergent subsequence whose limit belongs to the space.*

Proposition. *If $E \subset X$ is compact, then E is closed and bounded.*

Proof. Closed: If $(x_n) \subset E$ and $x_n \rightarrow x$ in X , since E is compact there exists a subsequence $x_{n_k} \rightarrow y$ with $y \in E$. But convergence in metric spaces is unique, hence $x = y \in E$. Thus E contains the limits of its sequences and is closed.

Bounded: Fix $x_0 \in X$. The balls $\{B(x_0, n)\}_{n \in \mathbb{N}}$ cover X , hence they cover E . By compactness, there exists N such that $E \subset B(x_0, N)$. This shows that E is bounded. $\square$

Example (Closed and bounded but not compact in the discrete metric). Consider $X = \{1/n : n \in \mathbb{N}\}$ with the discrete metric d_{disc} . Then every subset (in particular each singleton $\{1/n\}$) is open. The family $\mathcal{U} = \{\{1/n\} : n \in \mathbb{N}\}$ is an open cover of X with no finite subcover, so X is not compact. Nevertheless, X is bounded and closed in itself.

Theorem 3 (Heine–Borel). *In $(\mathbb{R}^j, d_{\text{euc}})$ a set $E \subset \mathbb{R}^j$ is compact if and only if it is closed and bounded.*

Proposition (Maximum in compact subsets of $\mathbb{R}$). *Let $E \subseteq \mathbb{R}$ be nonempty and compact. Then E has a maximum and a minimum.*

Proof. By Proposition, E is closed and bounded; in particular it has a supremum $\bar{x} = \sup E \in \mathbb{R}$. For each n there is $x_n \in E$ with $\bar{x} - \frac{1}{n} < x_n \leq \bar{x}$. Then $x_n \rightarrow \bar{x}$; since E is closed, $\bar{x} \in E$, and therefore $\max E = \bar{x}$. $\square$

Proposition. *Let (X, d) be a compact metric space and $F \subset X$ a closed subset. Then F is compact.*

Proof. If $\{U_i\}_{i \in I}$ is an open cover of F , then $\{U_i\}_{i \in I} \cup \{X \setminus F\}$ is an open cover of X . By compactness of X it admits a finite subcover; removing (if it appears) $X \setminus F$, we obtain a finite subcover of F . $\square$

Proposition. *Every compact metric space is complete.*

Proof. Let (x_n) be a Cauchy sequence in compact X . Then it has a convergent subsequence $x_{n_k} \rightarrow x \in X$. Since (x_n) is Cauchy, every subsequence has the same Cauchy bound and necessarily $x_n \rightarrow x$. $\square$

Continuity in metric spaces

Fix metric spaces (X, d_X) and (Y, d_Y) and a function $f : X \rightarrow Y$.

Definition (Continuity at a point). We say that f is continuous at $x_0 \in X$ if for every $\varepsilon > 0$ there exists $\delta > 0$ such that

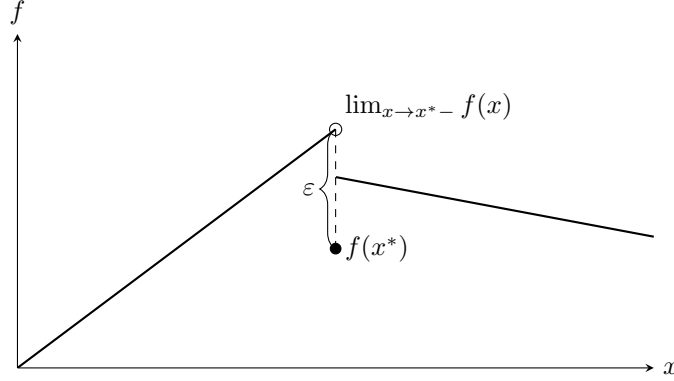
$$d_X(x, x_0) < \delta \implies d_Y(f(x), f(x_0)) < \varepsilon.$$

We say that f is continuous if it is continuous at every point of X .

Proposition (Sequential characterization). *f is continuous at x_0 if and only if for every sequence $(x_n) \subset X$ with $x_n \rightarrow x_0$ we have $f(x_n) \rightarrow f(x_0)$ in Y .*

Continuity

Let (X, d_X) and (Y, d_Y) be two metric spaces and $f : X \rightarrow Y$ a function. Intuitively, a *discontinuity* at x^* occurs if there are points x' near x^* such that $f(x')$ is not near $f(x^*)$.



Definition (Pointwise continuity). We say that $f : X \rightarrow Y$ is *continuous at* $x \in X$ if for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$d_X(x, x') < \delta \implies d_Y(f(x), f(x')) < \varepsilon.$$

The function is *continuous* if it is continuous at every $x \in X$.

Proposition (Sequential characterization). f is continuous at $x \in X$ if and only if for every sequence $(x_n)_{n \in \mathbb{N}}$ in X with $x_n \rightarrow x$ we have

$$f(x_n) \longrightarrow f(x) \quad \text{in } Y.$$

Proposition (Topological characterization). A function $f : X \rightarrow Y$ is continuous if and only if for every open set $U \subseteq Y$ the preimage $f^{-1}(U) = \{x \in X : f(x) \in U\}$ is open in X . The same is true for closed sets.

Example (Basic examples of continuous functions). 1) **Identity**. If $X = Y$ and $f(x) = x$, then f is continuous (indeed, $f^{-1}(U) = U$).

2) **Constant function**. If $f(x) = y_0$ for all $x \in X$, then f is continuous (the preimage of any open $U \subseteq Y$ is X if $y_0 \in U$ and $\emptyset$ otherwise).

3) **Finite domain**. If X is finite (with any metric), every subset of X is open; therefore, *every* function $f : X \rightarrow Y$ is continuous.

4) **Linear functions in $\mathbb{R}$** . With the usual metric $d(u, v) = |u - v|$, the function $f(x) = ax + b$ ($a \neq 0$) is continuous. Given $\varepsilon > 0$, choose $\delta = \varepsilon/|a|$, since

$$|x - x'| < \delta \implies |f(x) - f(x')| = |a||x - x'| < |a|\delta = \varepsilon.$$

Proposition (Algebra of continuous functions). Let $f, g : X \rightarrow \mathbb{R}$ be continuous (with the usual metric on $\mathbb{R}$). Then the following are also continuous:

- (i) $f + g$;
- (ii) $f \cdot g$;
- (iii) $\frac{f}{g}$, provided $g(x) \neq 0$ for all $x \in X$;
- (iv) $\max\{f, g\}$;
- (v) $\min\{f, g\}$;
- (vi) $|f| : x \mapsto |f(x)|$.

Definition (Uniform continuity). We say that $f : X \rightarrow Y$ is *uniformly continuous* if for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$d_X(x, x') < \delta \implies d_Y(f(x), f(x')) < \varepsilon \quad \text{for all } x, x' \in X.$$

The key difference with pointwise continuity is that here δ does not depend on the point x .

Example (Not uniformly continuous in $\mathbb{R}$). Let $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x^2$, with the usual metric. f is continuous on $\mathbb{R}$, but *not* uniformly continuous.

Proof. Take $\varepsilon = 1$. Let $\delta > 0$ be arbitrary and define

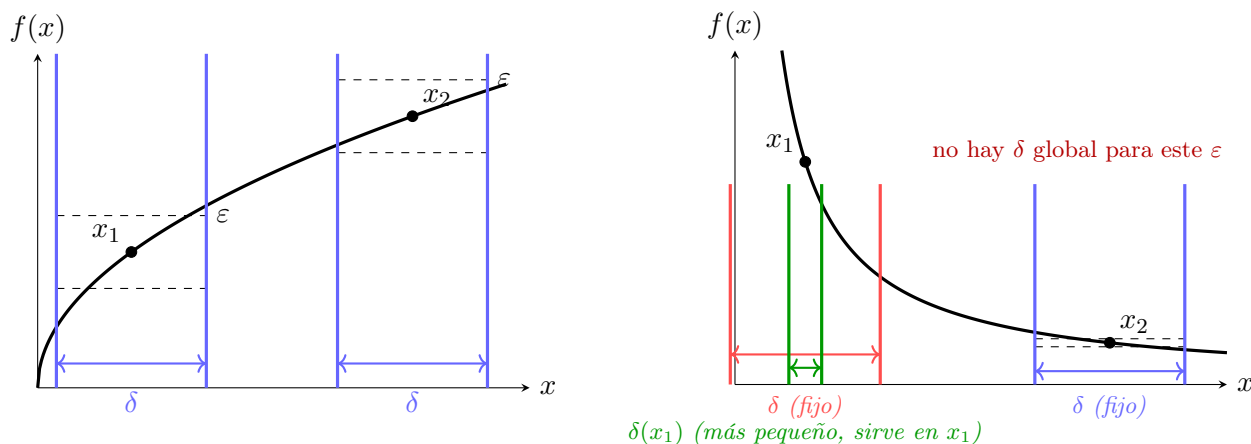
$$x := \frac{1}{\delta} + \frac{\delta}{4}, \quad x' := x - \frac{\delta}{2}.$$

Then $|x - x'| = \delta/2 < \delta$, but

$$|x^2 - (x')^2| = |x - x'| \cdot |x + x'| = \frac{\delta}{2} \left(2x - \frac{\delta}{2} \right) = \delta x - \frac{\delta^2}{4} = \delta \left(\frac{1}{\delta} + \frac{\delta}{4} \right) - \frac{\delta^2}{4} = 1.$$

In particular, for that pair x, x' we have $|x^2 - (x')^2| \geq \varepsilon$ even though $|x - x'| < \delta$. Since this holds for every $\delta > 0$, f is not uniformly continuous. $\square$

(a) **Uniform continuity:** $f(x) = \sqrt{x}$ en $[0, 1]$. Mismo δ funciona para cualquier x_0 (dados ε).
 (b) **Pointwise continuity:** $f(x) = 1/x$ en $(0, 1]$. El δ requerido depende fuertemente de x_0 .



Lets build intuition. We have to different but related concepts:

- **Uniform continuity.** Given a function $f : A \rightarrow \mathbb{R}$, we say it is uniformly continuous if for every $\varepsilon > 0$ there exists a *single* $\delta > 0$ (which does not depend on the point in the domain) such that, for any $x, y \in A$,

$$|x - y| < \delta \implies |f(x) - f(y)| < \varepsilon.$$

The “uniformity” means precisely that *one* $\delta(\varepsilon)$ works *across the whole domain*. Geometrically: imagine a “caliper” that opens δ along the x -axis. If, while sliding it over the graph, the images of any pair of points separated by less than δ always remain within a vertical band of height ε , then f is uniformly continuous. Choosing δ amounts to ensuring that this caliper works in the *worst corner* of the domain; if it works there, it will surely work elsewhere.

- **Pointwise (or “conventional”) continuity.** In contrast, f is continuous at x_0 if for every $\varepsilon > 0$ there exists $\delta = \delta(\varepsilon, x_0)$ such that

$$|x - x_0| < \delta \implies |f(x) - f(x_0)| < \varepsilon.$$

Here δ may depend on the point x_0 . That is why a function may be continuous at every point and yet *fail* to be uniformly continuous: the required δ keeps shrinking “more and more” as we move across certain regions of the domain, and there is *no single* δ that works for all points simultaneously.